Exact curve-cover transfer assessment

Current-catalog crosswalk for two explicit hyperelliptic covers

2026-10-13

Outcome

Two exact same-field hyperelliptic covers are constructively and reproducibly identified. Only one has any implemented Jacobian transfer, and that transfer is deliberately bounded.

- The binary degree-three map over $\mathbf{F}_{2^7}$ has verified curve geometry, checked one-infinity genus-three Jacobian arithmetic, pullback of rational elliptic point classes, pushforward of rational source-point divisors and the implementation's own typed fibers, and a narrow order-29 subgroup certificate.
- Its arbitrary-class $\mathrm{Jac}(H) \to E$ norm is **not** implemented. The public negative-control path reports it as unsupported.
- The prime degree-two map over $\mathbf{F}_{827}$ has verified genus-two cover geometry. Its source has two rational infinity points, while the current prime Jacobian accepts only odd-degree, one-infinity models. Prime-cover Jacobian arithmetic and induced maps therefore remain absent.
- No measured computational advantage, complete Jacobian relation solver, linear algebra, or target decomposition is claimed.

The canonical machine assessment is `docs/curves/cover-transfer-assessments.json`, validated by `cover-transfer-assessment.schema.json`. Its SHA-256 is:

`edd6101f4a9339d14ac02c259f7c323a718da6dee7918097f1053f016b4db9a8`

The semantic verifier independently pins this complete digest, the catalog base commit, and the current implementation and contract sources used below.

All claim states use exactly `unknown`, `supported`, `refuted`, and `not_applicable`. `Refuted` always names a precise implementation or evidence claim; it is never used to infer mathematical nonexistence from a failed or missing construction.

Why this is a continuation, not a rewrite

The October 6 infrastructure audit is an immutable, internally consistent snapshot. It pins 121 cover rows, two rich evidence records, exact source revisions, and the primary-source theorem audit. Rewriting its counts would destroy provenance. This continuation instead joins those same two exact records to the current HC1/CV1 graph and pins the current catalog inputs.

The pinned binary transfer, binary Jacobian, and prime curve kernels still match that snapshot. The catalog checker and standalone CLI have since changed, so the sidecar marks wholesale freshness of the historical replay `unknown` and uses current graph joins as separate evidence.

The primary-source conclusion remains:

- Xarles, *Hyperelliptic curves covering an elliptic curve twice*, arXiv:1303.4220v1, Corollary 3, works over an arbitrary field K of characteristic different from 2 and 3.
- It gives a hyperelliptic H/K of genus at most 5 and two independent K-defined maps $H \to E$. For $j \neq 0, 1728$, the displayed genus-five construction has two degree-three maps, hence separable in the stated characteristic. The exceptional j-cases cite genus-two work.

- This is not a uniform executable construction for every input model, and it is not the trivial genus-one interpretation. The standard hyperelliptic usage here has genus at least two.

The repository's fixed prime and binary constructors are separate, machine-replayed families. They establish neither all cases of Xarles's two-map theorem nor a solver.

Current catalog snapshot

Assessment base commit:

`6600c7c6c0cfbb6a9e4e596bf8c5cfa292fb269e`

Artifact	SHA-256	Audited content
`registry.json`	`9e420a7f...19e872`	2,091 ICV1 cards
`covers.json`	`baac4c80...928324`	2,091 verified geometry rows
`cover-links.yaml`	`a662b445...8e145`	2,091 maps, 2,089 unique covers, 664 EC1 representations
`jacobian-transfers.json`	`aec63380...e6ca40`	stale 321-curve formula screen; not imported as certificates
prior `evidence.json`	`b5860bea...9dff6`	two detailed exact records

Full artifact digests:

```
registry.json            9e420a7f2e69a1d72a2590899e6bffb8b3d45eff0f197c5c898ee66c1219e872
covers.json              baac4c80f89e3cd9c8ef14421b600cce98c9a642b54ed1f182e8dcc8a2928324
cover-links.yaml         a662b4458eab6d6c10fc7a51b74c0d2c7f6de3a9f817ea981ee898ada8f8e145
jacobian-transfers.json  aec6338011158c8c1aaddb829d4e90fb96eebc43cd4027d1067979bba3e6ca40
prior evidence.json      b5860beae57838245146c5183aa1a96a5c2bf78f2f1b7c669fb401cc8439dff6
```

Every `covers.json` row is `verified_over_declared_field`, every row still has `subgroup_transfer: not_tested`, and every `dlp_advantage` is null. The content-addressed graph likewise has 664 `not_tested` subgroup-transport entries. This assessment crosswalks two maps and two EC1 representations; the other 2,089 map rows and 662 EC1 representations remain unassessed for transfer purposes.

The separate `jacobian-transfers.json` file pins registry digest `f8297bb8...`, contains only 321 curves and 1,156 formula routes, and omits 1,770 current slugs. Its conditional labels and modeled formulas are useful discovery leads, not cover-map, kernel, image-order, or recovery certificates.

Browser impact was reviewed. This change adds no curve, map, solver result, or performance datum and deliberately leaves the generated lab browser and its geometry-only cards unchanged. The curves index links the new sidecar and report; a future browser transfer badge must consume these four-state claims rather than infer transfer from `verified_over_declared_field`.

Typed correspondence graph

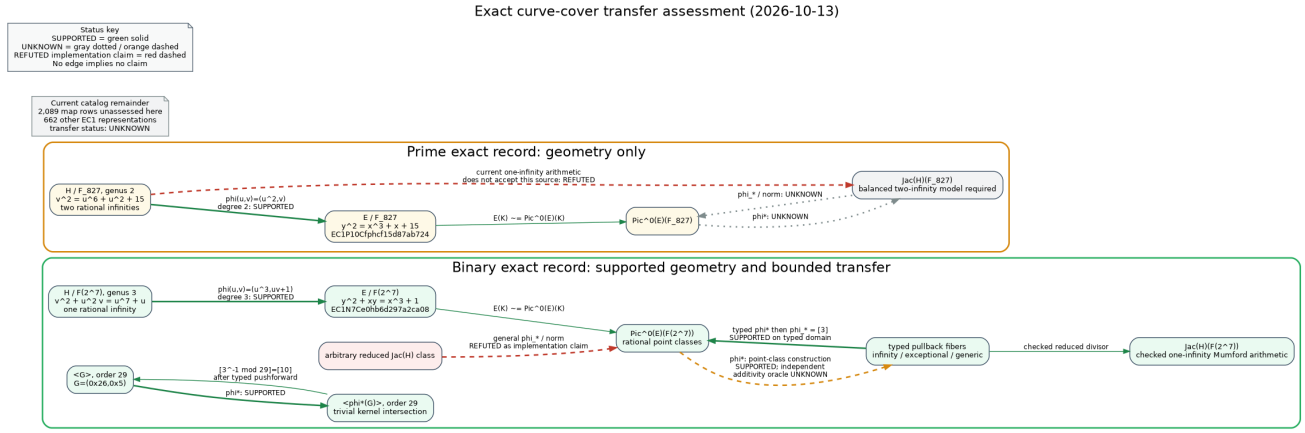


Figure 1: Typed correspondence graph.

Editable source: `typed_correspondence.dot`. Vector rendering: `typed_correspondence.svg`.

Green edges are supported only on their labels' exact domains. The binary pullback route is orange because its point-class constructor is supported while an independent additivity oracle remains unknown. The red binary norm edge refutes the claim that the current API handles arbitrary Jacobian classes. The red prime arithmetic edge refutes only compatibility with the existing one-infinity implementation; it says nothing against mathematical existence of a balanced Jacobian representation.

Exact records

Item	Binary record	Prime record
Field	$\mathbf{F}_2[z]/(z^7+z+1)$	$\mathbf{F}_{827}$
ICV1	`icv1-f2m7-t13-616700dd`	`icv1-fp10-t5-192cb216`
EC1	`EC1N7Ce0hb6d297a2ca08`	`EC1P10Cfphcf15d87ab724`
Target E	$y^2+xy=x^3+1$	$y^2=x^3+x+15$
Source H	$v^2+u^2v=u^7+u$	$v^2=u^6+u^2+15$
Genus / infinity	3 / one rational point	2 / two rational points
Map	$(u,v)\mapsto(u^3,uv+1)$	$(u,v)\mapsto(u^2,v)$
Degree / field	3 / same field	2 / same field
Cover ID	`HC1G3h2530bd70db89`	`HC1G2h998582908336`
Map ID	`CV1D3h0dc60016b03d`	`CV1D2h5768ce11909c`

The exact unabridged UIDs, model hashes, source models, target objects, subgroup generators, obligations, and evidence references are retained in the machine sidecar.

Transfer obligations

ID	Typed claim	Status	Exact boundary or remaining work
T1	Binary source, target, and degree-three morphism exist over the declared field	`supported`	Current HC1/CV1 objects and native polynomial replay

ID	Typed claim	Status	Exact boundary or remaining work
T2	Binary source smoothness, genus, separability, defining identity, infinity, and exceptional point	`supported`	Fixed-family verifier; see `cover_transfer.rs`
T3	Checked binary Jacobian validity, reduction, equality, negation, addition, and scalar multiplication	`supported`	One-infinity model only; implementation in `hyperelliptic.rs`
T4	Binary pullback ϕ^*	`supported`	Rational elliptic point classes only; infinity and exceptional fibers explicit
T5	Independent pullback additivity oracle on the exact $\mathbf{F}_{2^7}$ record	`unknown`	Internal consistency is not an independent CAS/divisor oracle
T6	Binary pushforward ϕ_*	`supported`	Single rational source-point divisors and private typed pullback witnesses only
T7	Arbitrary reduced-class binary norm	`refuted`	Public API returns a typed unsupported result; see `cover_transfer.rs`
T8	$\phi_*\phi^* = [3]$	`supported`	Typed rational point-class domain, exhaustively replayed across ordinary F8 models
T9	Exact order-29 kernel, image order, and degree-inverse recovery	`supported`	Only `EC1N7Ce0hb6d297a2ca08`; not a catalog-wide result
T10	Prime source, target, and degree-two morphism exist over $\mathbf{F}_{827}$	`supported`	Current HC1/CV1 objects and native geometry replay
T11	Current prime one-infinity arithmetic accepts the even sextic source	`refuted`	It requires degree $2g+1$; this source has degree $2g+2$ and two infinities; see `curve.rs`
T12	Prime pullback, norm, and $\phi_*\phi^* = [2]$	`unknown`	Balanced divisor representation and induced maps are not implemented
T13	Order-823 subgroup preservation for the prime record	`unknown`	Exact EC1 identity exists; kernel, image, and recovery evidence does not
T14	The remaining current cover-map rows preserve their catalog subgroups	`unknown`	Absence of assessment is not refutation
T15	Any cover here yields a computational advantage	`unknown`	No full pipeline or matched end-to-end measurement
T16	A failed bounded constructor proves nonexistence	`refuted`	Structured unsupported/search failure never proves no cover exists

For a degree-d finite separable morphism, the expected identity is $\phi_*\phi^* = [d]$ on $\mathrm{Jac}(E)$. Even a verified identity does not by itself prove preservation of a selected subgroup. The kernel intersection, exact image order, and recovery route must be bound to the full EC1 object. The order-29 fixture does this only for its narrow typed domain.

Cost and benefit assessment

No calibrated benchmark was performed. Correctness-test elapsed time is not an algorithmic cost, and a symbolic formula is not a measurement.

Phase	Value	Status	Reason
Cover construction	null	`unknown`	executable but uncalibrated
Failed searches	null	`unknown`	no catalog-wide search ledger
Duplicate candidates	null	`unknown`	no deduplication ledger
Geometry verification	null	`unknown`	replayable but uncalibrated
Jacobian arithmetic	null	`unknown`	correctness receipts only
Pullback / pushforward	null	`unknown`	bounded implementation, no measured conversion cost
Subgroup recovery	null	`unknown`	order-29 correctness witness only
Reusable preprocessing	null	`unknown`	no amortization boundary
Peak memory	null	`unknown`	not measured
Total runtime	null	`unknown`	no end-to-end transfer or solver run
Matched generic baseline	null	`unknown`	not run
Jacobian relations / linear algebra / target decomposition	null	`not_applicable`	separate solver was not implemented or run
End-to-end speedup	null	`unknown`	no complete numerator or denominator

Benefit hypothesis: a cover helps only if a selected EC1 subgroup embeds in an operationally usable source-Jacobian subgroup and the complete destination algorithm plus both conversions is cheaper than the matched generic baseline. This assessment neither confirms nor measures that hypothesis.

Falsification condition: any missing general norm, failed kernel/image certificate, failed independent relation replay, or total measured cost at or above the matched baseline rejects the claimed end-to-end benefit even when the curve morphism itself is correct.

Controls

Control	Status	Evidence boundary
ICV1, EC1, HC1, and CV1 identity replay	`supported`	exact current hashes and UIDs
Cover geometry replay	`supported`	both fixed families
Binary arithmetic internal exhaustive control	`supported`	all 486 reduced F8 fixture classes and shared-support cases

Control	Status	Evidence boundary
Independent CAS/divisor-class operation oracle	`unknown`	not run
Transported-data composition	`supported`	binary typed rational point classes only
Coordinate-isomorphism commutation	`unknown`	not run
Exact subgroup kernel/image/recovery	`supported`	binary order-29 record only
Prime order-823 transfer	`unknown`	no induced maps
Malformed/tampered/unsupported negative controls	`supported`	structured invalid or unsupported results
Cross-repository source parity	`supported`	PR #377 merge content, pinned manifest, and native replay; not fresh exact-tip CI
Paired performance control	`not_applicable`	no benefit experiment occurred

The mirrored implementation in `aburan28/cryptanalysis` is merged through PR #377. The replay receipt pins merge `eadd838a...`, the parity manifest digest, the no-diff path set, and a successful native parity replay against this checkout. Its prior `fb-refs` content defect was repaired by PR #352; a descendant validator run recorded zero problems. Current exact-tip workflow freshness is still absent because the repository's Actions allowlist blocks setup and the workflow is manually disabled. This is reported as a CI-delivery gap, not as a mathematical failure.

Standalone release replay

Crypto PR #1492 merged as `7636a3c8e12c210c4b7ab611c791d1979a177d9e`. Its immediate release run 999 failed on Windows and did not create tag `v0.1.0-main.999`. A later descendant release run 1011 completed all eight jobs and published Linux GNU, Linux musl, macOS, and Windows Rust bundles containing `hyperelliptic-cover`.

The later descendant `v0.1.0-main.1371` was independently replayed for this assessment. The exact commands, inputs, stdout hashes, and catalog comparison are retained in the `release_replay.txt` receipt:

- tag commit `bbd330a9d8e9bb59346bb9f5f45416851d2af3db` contains the cover merge;
- the exact asset is `crypto-v0.1.0-main.1371-x86_64-unknown-linux-gnu.tar.gz`;
- its SHA-256 is `77e69e60a68ada60c7b5fe13503e326f50fce4594b96340eca402041e60dcb2d`, matching the published `SHA256SUMS`;
- the archive contains `hyperelliptic-cover`, `registry.json`, `covers.json`, and `cover-links.yaml`;
- direct `construct`, `verify`, and binary-Jacobian `identity` commands returned status `ok`; and
- `catalog-export` completed without `cargo run`. Its mathematical rows matched the bundled catalog; the exported file intentionally changed the producer provenance fields.

The latest current-main release attempt has unrelated native/isogeny job failures, while all three Rust bundle jobs (Linux, macOS, Windows) passed. That newer failure does not erase the published descendant bundles above.

Exploration boundary

This work did not search production targets, recover keys, or run a discrete logarithm campaign. It assessed supplied constructions and current repository evidence only.

It does not:

- search all cover families, genera, degrees, or characteristics;
- assess the other 2,089 current map rows beyond their existing geometry status;
- turn missing evidence or failed bounded search into nonexistence;
- import stale formula screens as transfer certificates;
- claim a general binary norm or any prime induced map;

- claim subgroup preservation without exact EC1 evidence; or
- implement the full Jacobian index-calculus solver.

Open obligations

1. Implement and independently verify an arbitrary-class binary norm.
2. Replay pullback additivity and full composition through a genuinely independent divisor-class implementation.
3. Add balanced/two-infinity prime Jacobian arithmetic and prime-family induced maps.
4. Bind kernel, image order, and recovery certificates to every EC1 subgroup for which transfer is claimed.
5. Add coordinate-isomorphism controls.
6. Refresh the formula-screen inventory against the current registry without presenting modeled routes as certificates or measurements.
7. Assess the remaining 2,089 maps and 662 EC1 representations explicitly.
8. Keep the full solver separate. Its acceptance gate must require actual Jacobian relations, independently verified linear algebra and target decomposition, checked outputs, explicit parameter limits, and complete preprocessing, failure, duplicate, memory, and total-runtime accounting.

Reproduction

Structural and identity checks:

```
jq -e . docs/curves/cover-transfer-assessment.schema.json
jq -e . docs/curves/cover-transfer-assessments.json
jsonschema \
  -i docs/curves/cover-transfer-assessments.json \
  docs/curves/cover-transfer-assessment.schema.json
sha256sum docs/curves/registry.json docs/curves/covers.json \
  docs/curves/cover-links.yaml \
  docs/curves/jacobian-transfers.json
node research/cover_transfer_assessment_20261013/verify_assessment.mjs
```

Visual and report derivation:

```
dot -Tsvg typed_correspondence.dot -o typed_correspondence.svg
dot -Tpng typed_correspondence.dot -o typed_correspondence.png
pandoc README.md \
  --from=markdown+tex_math_single_backslash \
  --pdf-engine=pdflatex \
  -o cover_transfer_assessment_20261013.pdf
pdfinfo cover_transfer_assessment_20261013.pdf
```

Relevant sources are the binary transfer, binary Jacobian, prime Jacobian, and catalog checker, plus the CLI documentation and prior detailed evidence.